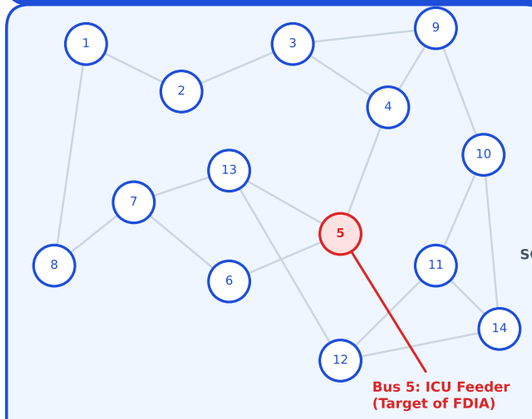
An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

1. Hospital Microgrid & OT SCADA



Jacobian Stealth Attack: $a = Hc$

χ^2 Bad-Data Residual: $r \leq \tau_{\chi^2}$ (Bypassed ✗)

SCADA Telemetry: $z = [V, \theta, P, Q]^T$

Clinical Loads: Ventilators • Cold Chain

IEC 60601-1 / IEEE 1547 Trip Limit: 100ms

IEC 61850 Substation Busbars • RTUs • PMUs

SCADA Telemetry →

2. Virtual Graph Digital Twin

Admittance Mirror: $G = (V, E, Y_{\text{bus}})$

State Sync: $x_{\text{DT}} \leftarrow x_{\text{DT}} + K_{\text{sync}}(z - h(x))$

Rolling Baseline: $\bar{z}_{\text{base}} = 0.95\bar{z} + 0.05z_{\text{clean}}$

GraphSAGE Layer 1: Admittance Aggregation

GraphSAGE Layer 2: Spatial Feature Concatenation

GraphSAGE Layer 3: Multi-Task Graph Embedding

Message: $m_{uv} = \text{ReLU}(W_e[h_u \| G_{uv} \| B_{uv}])$

Aggregation: $|Y_{uv}|/\sum|Y_{vw}|$ Circuit Coupling

Edge-Preserving • Topology Outage $(N-1)$ Invariant

On-Premises OT Privacy: Zero Cloud Exfiltration

Graph Embeddings →

3. Multi-Task Detection & XAI

Detection Head (Graph-Level)

Status: STEALTHY ATTACK DETECTED ✓

Det. F1: 98.25% ± 0.34% | AUC: 0.9993 | Latency: 0.908ms

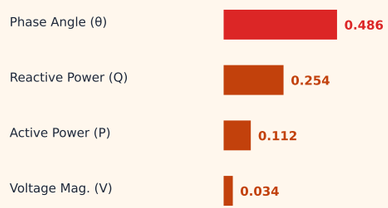
Localization Head (Node-Level)

Target Pinpointed: BUS 5 COMPROMISED ⚠

Loc. F1: 97.62% ± 0.84% | MCC: 0.982 (+5.52 pp via AG-MTL)

Loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + 1.5\mathcal{L}_{\text{WCE}} + 0.3\mathcal{L}_{\text{align}}$

Integrated Gradients Attribution on Bus 5:



Axiomatic Completeness: $\sum \text{Attr}_i = F(x) - F(x_0)$

Flagged Bus 5 →

4. Autonomous Closed-Loop Healing

Step 1: Automated RTU Quarantine

Isolate Bus 5 Corrupted SCADA Channels

Step 2: Admittance-Weighted Interpolation

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| z_j}{\sum |Y_{kj}|} + \delta$$

WLS State Estimator MSE Recovery:

Unmitigated Attack: 0.007228 p.u.²

Static Baseline: 0.003090 p.u.² (57.21%)

CGDT Reconstruction: 0.000002 p.u.² (99.97%)

Error Reduction: 99.97% | Cycle Time: < 2ms

Clinical Safety Compliance:

State Restored in < 2ms << 100ms Trip Limit

Ventilators & Surgical Suites Uninterrupted

Cross-Grid Inductive Scalability:

Outage Det. F1: 92.80% | Loc. F1: 86.96%

30-Bus Grid Transfer F1: 90.11% (Fine-Tuned)

◄ Autonomous RTU Quarantine & Admittance-Weighted Spatial Telemetry Reconstruction Injected into EMS (< 2ms) ◄